

```
In [1]: import pandas as pd
```

```
In [2]: pd.__version__
```

```
Out[2]: '2.3.3'
```

```
In [3]: df=pd.read_excel(r"C:\Users\madhu\Downloads\data.xlsx")
df
```

```
Out[3]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

```
In [4]: df.isna()
```

```
Out[4]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

```
In [5]: df.shape
```

```
Out[5]: (195, 5)
```

```
In [6]: df.isnull().sum()
```

```
Out[6]: CountryName      0
CountryCode      0
BirthRate        0
InternetUsers    0
IncomeGroup      0
dtype: int64
```

```
In [7]: df.isna().sum()
```

```
Out[7]: CountryName      0
CountryCode      0
BirthRate        0
InternetUsers    0
IncomeGroup      0
dtype: int64
```

```
In [8]: id(df)
```

```
Out[8]: 2674642940464
```

```
In [9]: df.columns
```

```
Out[9]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
              'IncomeGroup'],
              dtype='object')
```

```
In [10]: len(df.columns)
```

```
Out[10]: 5
```

```
In [11]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   CountryName     195 non-null   object
1   CountryCode     195 non-null   object
2   BirthRate       195 non-null   float64
3   InternetUsers   195 non-null   float64
4   IncomeGroup     195 non-null   object
dtypes: float64(2), object(3)
memory usage: 7.7+ KB
```

```
In [12]: df
```

```
Out[12]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

```
In [13]: df.head()
```

Out[13]:	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [14]: `df.tail()`

Out[14]:	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

In [15]: `df.head(3)`

Out[15]:	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income

In [16]: `df`

Out[16]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

```
In [17]: df[['CountryName', 'CountryCode']]
```

Out[17]:

	CountryName	CountryCode
0	Aruba	ABW
1	Afghanistan	AFG
2	Angola	AGO
3	Albania	ALB
4	United Arab Emirates	ARE
...	...	...
190	Yemen, Rep.	YEM
191	South Africa	ZAF
192	Congo, Dem. Rep.	COD
193	Zambia	ZMB
194	Zimbabwe	ZWE

195 rows × 2 columns

```
In [18]: df.columns
```

```
Out[18]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',  
              'IncomeGroup'],  
              dtype='object')
```

```
In [19]: df_cat=df[['CountryName', 'CountryCode', 'IncomeGroup']]  
df_cat
```

```
Out[19]:
```

	CountryName	CountryCode	IncomeGroup
0	Aruba	ABW	High income
1	Afghanistan	AFG	Low income
2	Angola	AGO	Upper middle income
3	Albania	ALB	Upper middle income
4	United Arab Emirates	ARE	High income
...	...	...	...
190	Yemen, Rep.	YEM	Lower middle income
191	South Africa	ZAF	Upper middle income
192	Congo, Dem. Rep.	COD	Low income
193	Zambia	ZMB	Lower middle income
194	Zimbabwe	ZWE	Low income

195 rows × 3 columns

```
In [20]: df_num=df[['BirthRate', 'InternetUsers', 'IncomeGroup']]  
df_num
```

Out[20]:

	BirthRate	InternetUsers	IncomeGroup
0	10.244	78.9	High income
1	35.253	5.9	Low income
2	45.985	19.1	Upper middle income
3	12.877	57.2	Upper middle income
4	11.044	88.0	High income
...	...	...	...
190	32.947	20.0	Lower middle income
191	20.850	46.5	Upper middle income
192	42.394	2.2	Low income
193	40.471	15.4	Lower middle income
194	35.715	18.5	Low income

195 rows × 3 columns

```
In [21]: print(df.shape)
print(df_num.columns)
print(df_cat.columns)
```

```
(195, 5)
Index(['BirthRate', 'InternetUsers', 'IncomeGroup'], dtype='object')
Index(['CountryName', 'CountryCode', 'IncomeGroup'], dtype='object')
```

```
In [22]: df.describe()
```

Out[22]:

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

```
In [23]: df.describe()
```

Out[23]:

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

In [24]: `df_cat.describe()`

Out[24]:

	CountryName	CountryCode	IncomeGroup
count	195	195	195
unique	195	195	4
top	Aruba	ABW	High income
freq	1	1	67

Indexing and slicing in Pandas

In [25]: `df`

Out[25]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [26]: df[:]

Out[26]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [27]: df[4:10]

Out[27]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
4	United Arab Emirates	ARE	11.044	88.0000	High income
5	Argentina	ARG	17.716	59.9000	High income
6	Armenia	ARM	13.308	41.9000	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.4000	High income
8	Australia	AUS	13.200	83.0000	High income
9	Austria	AUT	9.400	80.6188	High income

In [28]: df[0:150:10]

Out[28]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.900000	High income
10	Azerbaijan	AZE	18.300	58.700000	Upper middle income
20	Belarus	BLR	12.500	54.170000	Upper middle income
30	Canada	CAN	10.900	85.800000	High income
40	Costa Rica	CRI	15.022	45.960000	Upper middle income
50	Ecuador	ECU	21.070	40.353684	Upper middle income
60	Gabon	GAB	30.555	9.200000	Upper middle income
70	Greenland	GRL	14.500	65.800000	High income
80	India	IND	20.291	15.100000	Lower middle income
90	Kazakhstan	KAZ	22.730	54.000000	Upper middle income
100	Libya	LBY	21.425	16.500000	Upper middle income
110	Moldova	MDA	12.141	45.000000	Lower middle income
120	Mozambique	MOZ	39.705	5.400000	Low income
130	Netherlands	NLD	10.200	93.956400	High income
140	Poland	POL	9.600	62.849200	High income

In [29]: df[2:-1]

Out[29]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
5	Argentina	ARG	17.716	59.9	High income
6	Armenia	ARM	13.308	41.9	Lower middle income
...	...	...	...	...	...
189	Samoa	WSM	26.172	15.3	Lower middle income
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income

192 rows × 5 columns

In [30]: `df[::2]`

Out[30]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5	Low income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
188	West Bank and Gaza	PSE	30.394	46.6	Lower middle income
186	Vietnam	VNM	15.537	43.9	Lower middle income
...	...	...	...	...	...
8	Australia	AUS	13.200	83.0	High income
6	Armenia	ARM	13.308	41.9	Lower middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
2	Angola	AGO	45.985	19.1	Upper middle income
0	Aruba	ABW	10.244	78.9	High income

98 rows × 5 columns

In [31]: `df[::-10]`

Out[31]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5000	Low income
184	Venezuela, RB	VEN	19.842	54.9000	High income
174	Trinidad and Tobago	TTO	14.590	63.8000	High income
164	Swaziland	SWZ	30.093	24.7000	Lower middle income
154	Sierra Leone	SLE	36.729	1.7000	Low income
144	French Polynesia	PYF	16.393	56.8000	High income
134	Oman	OMN	20.419	66.4500	High income
124	Malaysia	MYS	16.805	66.9700	Upper middle income
114	Macedonia, FYR	MKD	11.222	65.2400	Upper middle income
104	Lesotho	LSO	28.738	5.0000	Lower middle income
94	Kiribati	KIR	29.044	11.5000	Lower middle income
84	Iceland	ISL	13.400	96.5468	High income
74	Hong Kong SAR, China	HKG	7.900	74.2000	High income
64	Guinea	GIN	37.337	1.6000	Low income
54	Estonia	EST	10.300	79.4000	High income
44	Czech Republic	CZE	10.200	74.1104	High income
34	Cote d'Ivoire	CIV	37.320	8.4000	Lower middle income
24	Brazil	BRA	14.931	51.0400	Upper middle income
14	Burkina Faso	BFA	40.551	9.1000	Low income
4	United Arab Emirates	ARE	11.044	88.0000	High income

In [32]: `df[:1]`

Out[32]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income

2nd Feb

In [33]: `df.columns`

Out[33]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
 'IncomeGroup'],
 dtype='object')

```
In [34]: df.columns=['a', 'b','c','d','e']
```

```
In [35]: df.head(2)
```

```
Out[35]:
```

	a	b	c	d	e
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income

```
In [36]: df.columns #column rename
```

```
Out[36]: Index(['a', 'b', 'c', 'd', 'e'], dtype='object')
```

```
In [50]: df.columns=['CountryName','CountryCode','BirthRate','InternetUsers','IncomeGroup']
```

```
In [51]: df.head(2)
```

```
Out[51]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income

```
In [39]: df['InternetUsers']
```

```
Out[39]:
```

	InternetUsers
0	78.9
1	5.9
2	19.1
3	57.2
4	88.0
...	...
190	20.0
191	46.5
192	2.2
193	15.4
194	18.5

195 rows × 1 columns

```
In [40]: df
```

Out[40]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [54]: `df.InternetUsers < 2`

Out[54]:

0	False
1	False
2	False
3	False
4	False
...	
190	False
191	False
192	False
193	False
194	False

Name: InternetUsers, Length: 195, dtype: bool

In [74]: `Filter = df['InternetUsers'] < 2`

In [43]: `Filter`

Out[43]:

InternetUsers	
0	False
1	False
2	False
3	False
4	False
...	...
190	False
191	False
192	False
193	False
194	False

195 rows × 1 columns

```
In [80]: result= df.loc[Filter & Filter2]
print(result.head())
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup \
11	Burundi	BDI	44.151	1.3	Low income
52	Eritrea	ERI	34.800	0.9	Low income
55	Ethiopia	ETH	32.925	1.9	Low income
64	Guinea	GIN	37.337	1.6	Low income
117	Myanmar	MMR	18.119	1.6	Lower middle income

	mycalc
11	57.3963
52	31.3200
55	62.5575
64	59.7392
117	28.9904

```
In [55]: print(df.isnull().sum())
print(df.info())
```

```

CountryName      0
CountryCode      0
BirthRate        0
InternetUsers    0
IncomeGroup      0
dtype: int64
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   CountryName     195 non-null   object
1   CountryCode     195 non-null   object
2   BirthRate       195 non-null   float64
3   InternetUsers   195 non-null   float64
4   IncomeGroup     195 non-null   object
dtypes: float64(2), object(3)
memory usage: 7.7+ KB
None

```

```
In [52]: print(df.columns.tolist())
```

```
['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers', 'IncomeGroup']
```

```
In [56]: df[df.BirthRate>40]
```

```
Out[56]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
2	Angola	AGO	45.985	19.1	Upper middle income
11	Burundi	BDI	44.151	1.3	Low income
14	Burkina Faso	BFA	40.551	9.1	Low income
65	Gambia, The	GMB	42.525	14.0	Low income
115	Mali	MLI	44.138	3.5	Low income
127	Niger	NER	49.661	1.7	Low income
128	Nigeria	NGA	40.045	38.0	Lower middle income
156	Somalia	SOM	43.891	1.5	Low income
167	Chad	TCD	45.745	2.3	Low income
178	Uganda	UGA	43.474	16.2	Low income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income

```
In [75]: Filter2=df['BirthRate']>2
```

```
In [48]: Filter2
```


Out[48]:

BirthRate	
0	True
1	True
2	True
3	True
4	True
...	...
190	True
191	True
192	True
193	True
194	True

195 rows × 1 columns

In [59]: `df[Filter2]`

Out[59]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [81]: `df[(df.BirthRate >40) & (df.InternetUsers <2)]`
`df.head()`

Out[81]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	mycalc
0	Aruba	ABW	10.244	78.9	High income	808.2516
1	Afghanistan	AFG	35.253	5.9	Low income	207.9927
2	Angola	AGO	45.985	19.1	Upper middle income	878.3135
3	Albania	ALB	12.877	57.2	Upper middle income	736.5644
4	United Arab Emirates	ARE	11.044	88.0	High income	971.8720

In [72]: `print(df.index)`

RangeIndex(start=0, stop=195, step=1)

In [96]: `df['mycalc']=df.BirthRate*df.InternetUsers`
`df`

Out[96]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	mycalc
0	Aruba	ABW	10.244	78.9	High income	808.2516
1	Afghanistan	AFG	35.253	5.9	Low income	207.9927
2	Angola	AGO	45.985	19.1	Upper middle income	878.3135
3	Albania	ALB	12.877	57.2	Upper middle income	736.5644
4	United Arab Emirates	ARE	11.044	88.0	High income	971.8720
...	...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income	658.9400
191	South Africa	ZAF	20.850	46.5	Upper middle income	969.5250
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income	93.2668
193	Zambia	ZMB	40.471	15.4	Lower middle income	623.2534
194	Zimbabwe	ZWE	35.715	18.5	Low income	660.7275

195 rows × 6 columns

In [97]: `df.columns`

```
Out[97]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',  
              'IncomeGroup', 'mycalc'],  
             dtype='object')
```

```
In [98]: df[df.IncomeGroup=='Low income']
```

Out[98]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	mycalc
1	Afghanistan	AFG	35.253	5.90	Low income	207.99270
11	Burundi	BDI	44.151	1.30	Low income	57.39630
13	Benin	BEN	36.440	4.90	Low income	178.55600
14	Burkina Faso	BFA	40.551	9.10	Low income	369.01410
29	Central African Republic	CAF	34.076	3.50	Low income	119.26600
38	Comoros	COM	34.326	6.50	Low income	223.11900
52	Eritrea	ERI	34.800	0.90	Low income	31.32000
55	Ethiopia	ETH	32.925	1.90	Low income	62.55750
64	Guinea	GIN	37.337	1.60	Low income	59.73920
65	Gambia, The	GMB	42.525	14.00	Low income	595.35000
66	Guinea-Bissau	GNB	37.503	3.10	Low income	116.25930
77	Haiti	HTI	25.345	10.60	Low income	268.65700
93	Cambodia	KHM	24.462	6.80	Low income	166.34160
99	Liberia	LBR	35.521	3.20	Low income	113.66720
111	Madagascar	MDG	34.686	3.00	Low income	104.05800
115	Mali	MLI	44.138	3.50	Low income	154.48300
120	Mozambique	MOZ	39.705	5.40	Low income	214.40700
123	Malawi	MWI	39.459	5.05	Low income	199.26795
127	Niger	NER	49.661	1.70	Low income	84.42370
132	Nepal	NPL	20.923	13.30	Low income	278.27590
148	Rwanda	RWA	32.689	9.00	Low income	294.20100
154	Sierra Leone	SLE	36.729	1.70	Low income	62.43930
156	Somalia	SOM	43.891	1.50	Low income	65.83650
158	South Sudan	SSD	37.126	14.10	Low income	523.47660
167	Chad	TCD	45.745	2.30	Low income	105.21350
168	Togo	TGO	36.080	4.50	Low income	162.36000
177	Tanzania	TZA	39.518	4.40	Low income	173.87920
178	Uganda	UGA	43.474	16.20	Low income	704.27880
192	Congo, Dem. Rep.	COD	42.394	2.20	Low income	93.26680

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	mycalc
194	Zimbabwe	ZWE	35.715	18.50	Low income	660.72750

In [104... `df.IncomeGroup.unique()`

Out[104... `array(['High income', 'Low income', 'Upper middle income',
'Lower middle income'], dtype=object)`

In [99]: `df[df.IncomeGroup == 'High income']`

Out[99]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	mycalc
0	Aruba	ABW	10.244	78.90	High income	808.25160
4	United Arab Emirates	ARE	11.044	88.00	High income	971.87200
5	Argentina	ARG	17.716	59.90	High income	1061.18840
7	Antigua and Barbuda	ATG	16.447	63.40	High income	1042.73980
8	Australia	AUS	13.200	83.00	High income	1095.60000
...	...	...	...	...	...	...
174	Trinidad and Tobago	TTO	14.590	63.80	High income	930.84200
180	Uruguay	URY	14.374	57.69	High income	829.23606
181	United States	USA	12.500	84.20	High income	1052.50000
184	Venezuela, RB	VEN	19.842	54.90	High income	1089.32580
185	Virgin Islands (U.S.)	VIR	10.700	45.30	High income	484.71000

67 rows × 6 columns

In [103... `df.IncomeGroup.nunique()`

Out[103... `4`

In [102... `print(df.columns.to_list())`
`df=df.drop('mycalc', axis=1)`

```
['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers', 'IncomeGroup', 'mycal  
c']
```

	CountryName	CountryCode	BirthRate	InternetUsers	\
0	Aruba	ABW	10.244	78.9	
1	Afghanistan	AFG	35.253	5.9	
2	Angola	AGO	45.985	19.1	
3	Albania	ALB	12.877	57.2	
4	United Arab Emirates	ARE	11.044	88.0	
..	...	...	...	...	
190	Yemen, Rep.	YEM	32.947	20.0	
191	South Africa	ZAF	20.850	46.5	
192	Congo, Dem. Rep.	COD	42.394	2.2	
193	Zambia	ZMB	40.471	15.4	
194	Zimbabwe	ZWE	35.715	18.5	

	IncomeGroup
0	High income
1	Low income
2	Upper middle income
3	Upper middle income
4	High income
..	...
190	Lower middle income
191	Upper middle income
192	Low income
193	Lower middle income
194	Low income

[195 rows x 5 columns]

```
In [105... import matplotlib.pyplot as plt #visulaization
import seaborn as sns #distribution visualization
%matplotlib inline
plt.rcParams['figure.figsize']=6,2

import warnings
warnings.filterwarnings('ignore') #os error
```

```
In [106... df.head()
```

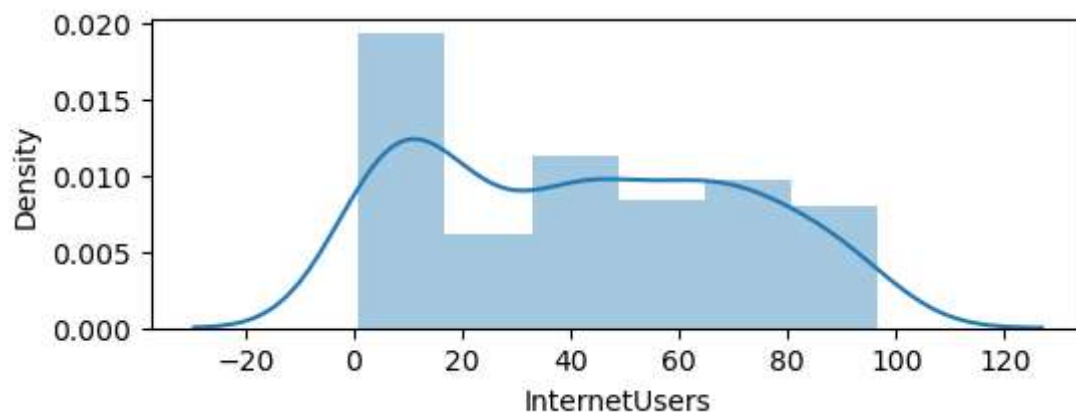
```
Out[106... CountryName CountryCode BirthRate InternetUsers IncomeGroup
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

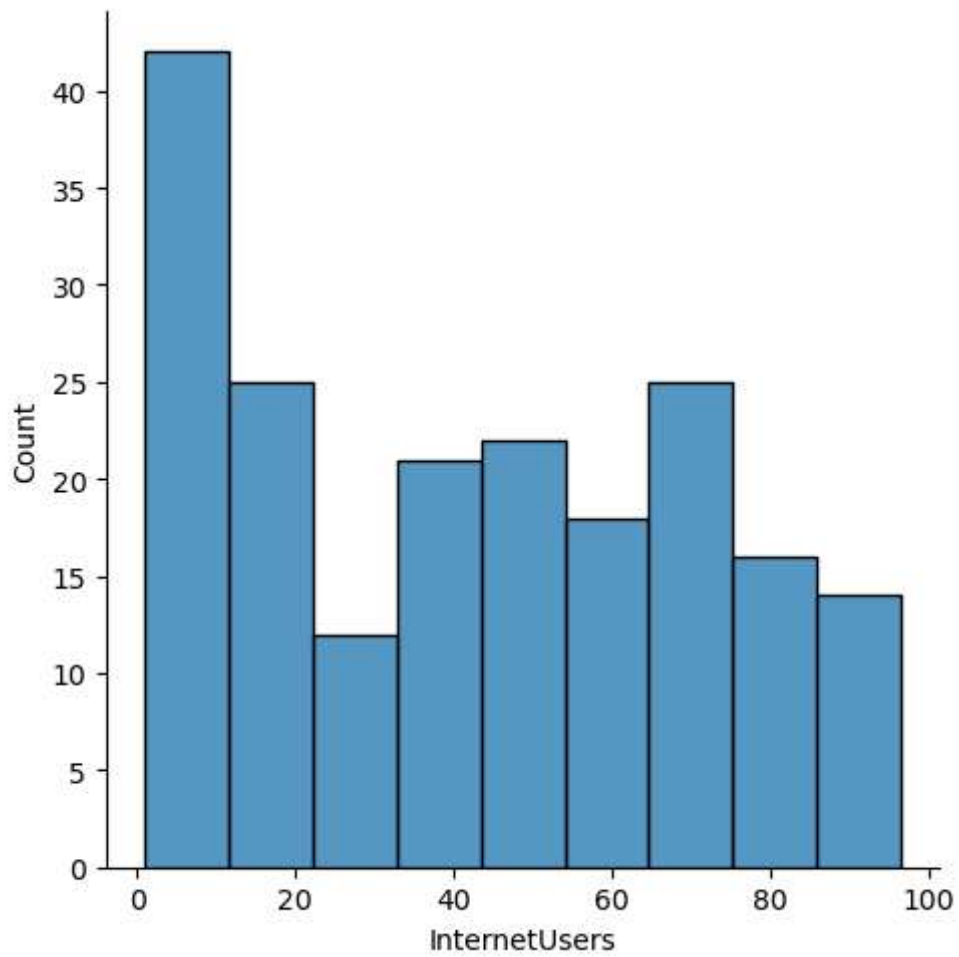
```
In [109... df["InternetUsers"]
```

```
Out[109...] 0      78.9
              1       5.9
              2      19.1
              3      57.2
              4      88.0
              ...
            190     20.0
            191     46.5
            192       2.2
            193     15.4
            194     18.5
Name: InternetUsers, Length: 195, dtype: float64
```

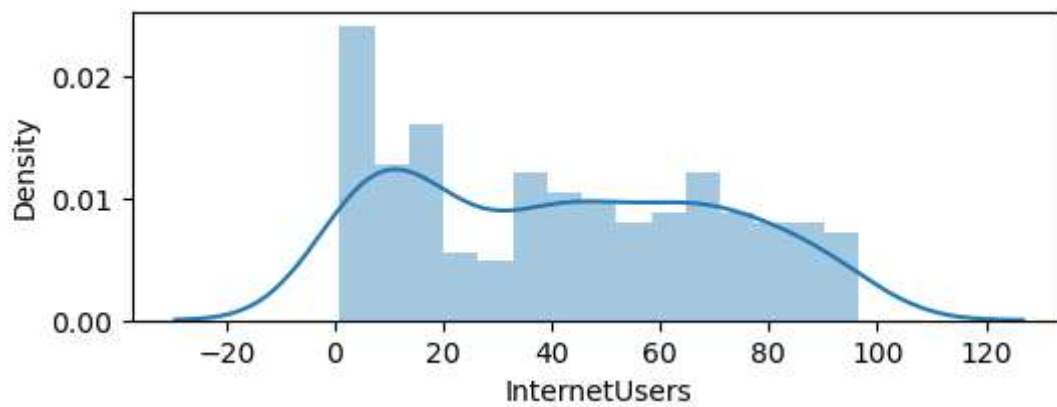
```
In [114...] vis1=sns.distplot(df["InternetUsers"])
              plt.show(vis1)
```



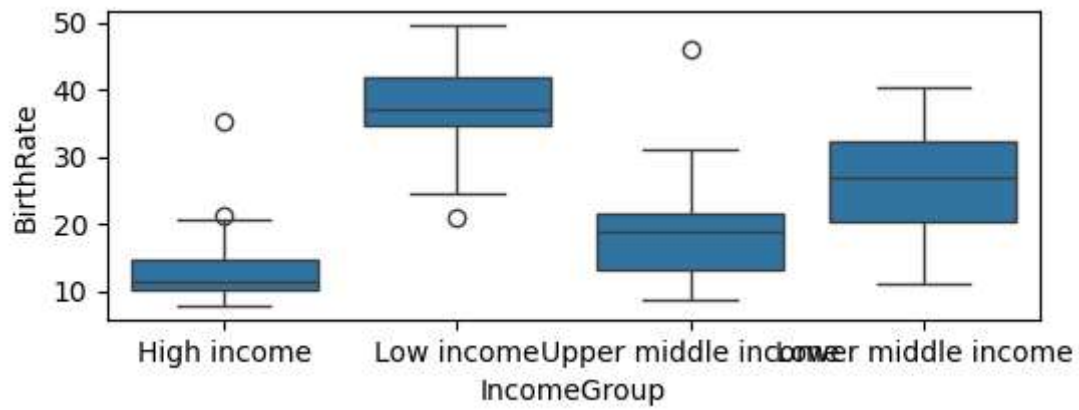
```
In [117...] vis2=sns.displot(df["InternetUsers"])
              plt.show()
```



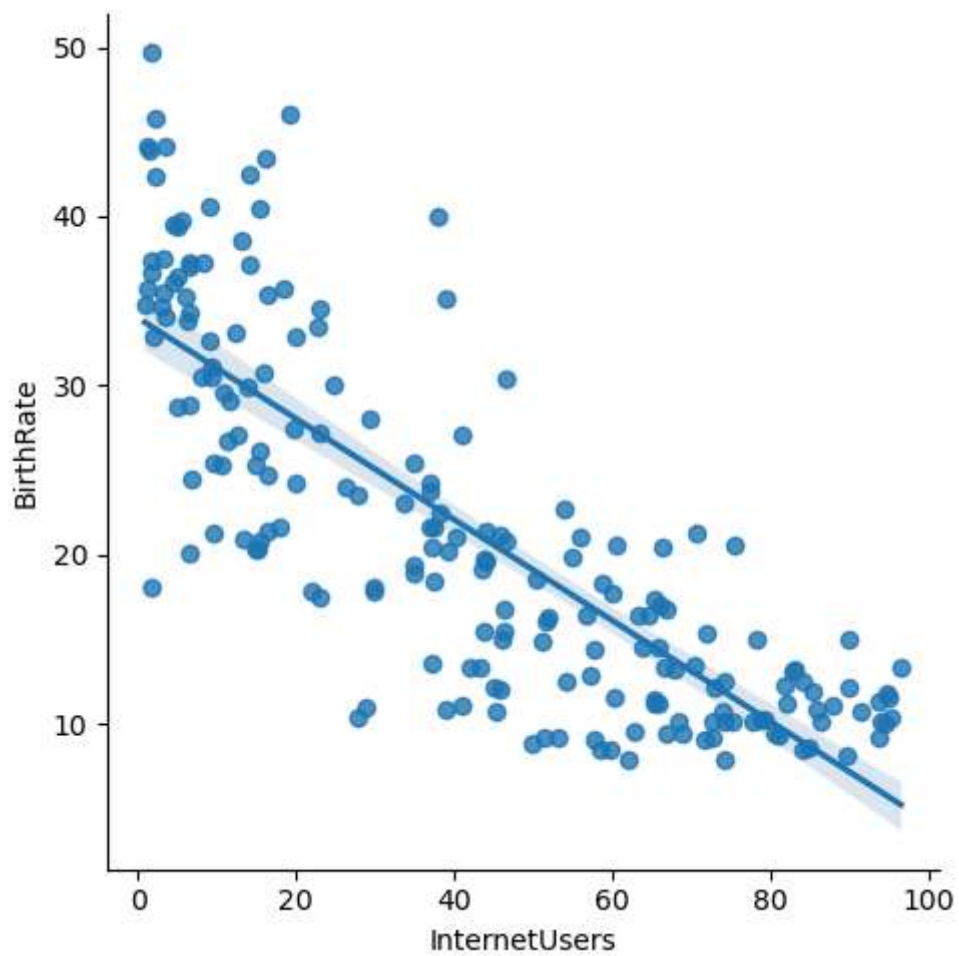
```
In [118... vis3=sns.distplot(df["InternetUsers"],bins=15)  
plt.show()
```



```
In [119... vis4=sns.boxplot(data=df, x="IncomeGroup", y="BirthRate")
```

```
In [121]: vis5 = sns.lmplot(data=df, x='InternetUsers', y="BirthRate")
plt.show()
```



```
In [ ]:
```